Marmara Üniversitesi
Mühendislik Fakültesi
Elektrik-Elektronik Mühendisliği

Laboratory Manual

EE 3011: Electronics I

Spring 2025

TABLE OF CONTENTS

Experiment Number	Experiment Title	Page	Comments/suggestions
0	Introductory Session. Laboratory regulations and safety rules	3	
1	Introduction to LT Spice (part 1)	4	
2	Introduction to LT Spice (part 2)	7	
3	Simulation of Diode Circuits using LT Spice	10	
4	Diode Characteristics	16	
5	Diode Rectifiers with Peak Detector	22	
6	MOSFET IV Characteristics	26	
7	The Common-source Amplifier	31	
8	BJT IV Characteristics	35	
-	Notes and Remarks	42	

LABORATORY REGULATIONS AND SAFETY RULES

Please observe the following regulations and safety rules inside the electric circuits laboratory at all times. These rules and regulations are prescribed to safeguard all laboratory users against electrical and other hazards.

1. It is your obligation to take all reasonable steps to safeguard the health and safety of yourself and other users of the laboratory. Also, equipment user is held responsible for any damage made to the equipment or other laboratory stuff, either directly or indirectly.
2. Make sure that the equipment you are planning to use are in working conditions. Immediately report any defective equipment/devices to the laboratory instructor/staff.
3. Use only the equipment provided in the experiment equipment list. If needed, ask the laboratory instructor for assistance.
4. Make sure to energize the power supply units in your circuits in the presence of the laboratory instructor.
5. Keep a safety distance between your working spot and any circuit breakers or any moving parts during the experiment.
6. Avoid making contact with energized circuits. A path to ground through your body can endanger your life.
7. After the completion of any experiment, make sure to switch off all equipment and disconnect the power supply units from the circuit before leaving the laboratory.
8. Observe tidiness and cleanliness and proper laboratory housekeeping of the equipment and other related accessories, including wires, cables, components, integrated circuits, etc. It is forbidden to make any writing and all kinds of marking on the equipment and the working place.
9. Pay a special attention to your clothing. Clothes with loose parts and/or poor protection should be avoided in the laboratory.
10. Before switching on the power supply unit, make sure you double-check the circuit connections.
11. Make sure that the last connection to be made in your circuit is the power supply and first thing to be disconnected is also the power supply.
12. Do not remove, transfer or relocate any laboratory equipment or other accessories without permission from the laboratory instructor/staff.
13. Make sure you have adequate training/orientation on a certain equipment prior to using it.
14. Smoking, drinks and food are not permitted in the laboratory.

Experiment (1)
Introduction to LTspice (Part 1)

Objectives

- To simulate simple passive circuits using DC and Transient analysis.
- To be able to generate LTspice results using plots for various circuit variables.

Pre-lab Work

Before starting this experiment, you need to study the document "Introduction to LTspice IV with Examples", by Nicholas Lombardo, Linear Technology, 2015.

Circuits

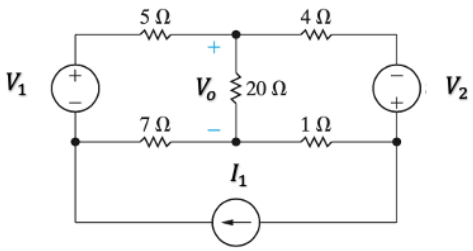


Figure 1: A DC circuit.

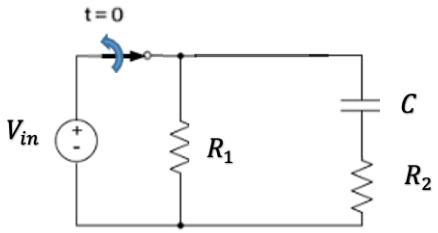


Figure 2: Transient circuit.

Procedure

Part (1): DC Analysis.

1. Connect the circuit in figure 1. Use the voltage and current sources values given by the lab instructor.
2. Using simulation, find the DC voltages and currents for all resistors. Record results in table 1.
3. Now make source V_1 as a variable voltage source and use the DC Sweep function to produce the transfer function curve V_o versus V_1.

Part (2): Transient Analysis.

1. Connect the circuit in figure 2. Use the circuit elements values given by the lab instructor. Set V_{in} as a periodic rectangular signal with frequency f and maximum amplitude V_{max}. The switch is initially closed.
2. Simulate the steady-state response of the circuit. Produce the steady-state voltage waveforms (versus time) across the capacitor C and the resistor R_2. Put these curves together with the V_{in} curve on the same plot.
3. Now, as the switch is opened at t=0, simulate the circuit using the Transient analysis function and generate the voltage waveform (versus time) across the capacitor C and the resistor R_2.

Put these curves together with the V_{in} curve on the same plot. Also observe the waveform of the current through R_2. From the curves, find the time constant of the circuit.

4. Change the value of f and repeat step 3. frekansı değiştir, v ın içinde time kısmında mesela 10m yap

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Results and Analysis

Part (1): DC Analysis.

Table 1. DC Simulations in Part (1).

$V_1 =$ 1V $V_2 =$ 2V $I_1 =$ 1A	Simulation	Calculation	% Difference
Voltage across 1 Ω resitor	-360.01 mV		
Current through 1 Ω resitor	-360.01 mA		
Voltage across 4 Ω resitor	-2.56 V		
Current through 4 Ω resitor	-640 mA		
Voltage across 5 Ω resitor	-3.250 V		
Current through 5 Ω resitor	-650 mA		
Voltage across 7 Ω resitor	-2.45 V		
Current through 7 Ω resitor	-350 mA		
Voltage across 20 Ω resitor	-200 mV		
Current through 20 Ω resitor	-10 mA		

1. Comment on the difference between simulation and theoretical values in table 1.

Part (2): Transient Analysis.

Table 2. Transient Simulations in Part (2).

R_1 =100 R_2 =10k C =1mikrof (u)	Simulation	Calculation	% Difference
Circuit time constant			

1. Comment on the difference between simulation and theoretical values in table 2.

2. State your observations on the simulated behavior of the given RC circuit, including the effect of changing the input frequency.

Experiment (2)
Introduction to LT Spice (Part 2)

Objectives

- To simulate a first-order low-pass filter circuit using AC analysis.
- To be able to present results using plots for various circuit variables.

Pre-lab Work

1) Before starting this experiment, you need to study the document "Introduction to LT Spice IV with Examples", by Nicholas Lombardo, Linear Technology, 2015.
2) Analyze the circuit in figure 1. In particular, find the frequency response of the transfer function and filter bandwidth.

Circuits

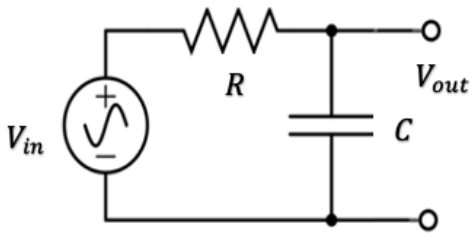


Figure 1: RC low-pass filter.

Procedure

Part (1): AC Analysis.

1. Connect the circuit in figure 1. Use the circuit elements values given by the lab instructor. V_{in} is a sinusoidal AC voltage source with frequency f and maximum amplitude V_{max}.
2. Simulate the circuit using the AC analysis function and generate the transfer function spectrum curve (versus frequency). Use dB scale for the transfer function magnitude and log scale for the frequency.
3. Change the value of C and observe the simulation curves.

Results and Analysis

Part (1): AC Analysis.

1. From your simulations, find the simulated bandwidth of the filter for different values of C. Record your results in table 1.

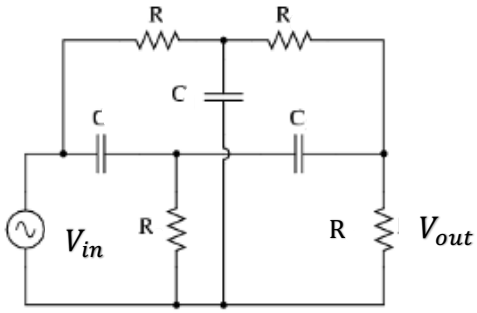
Table 1. Simulated and calculated bandwidth.

Capacitor value	Simulated Bandwidth	Theoretical Bandwidth	% Difference

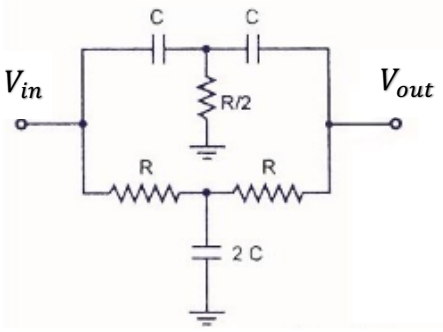
2. Comment on the results in table 1.

3. If the output of the circuit is taken across the resistor instead of the capacitor, what kind of a filter is obtained? Explain your answer and support it using simulation results.

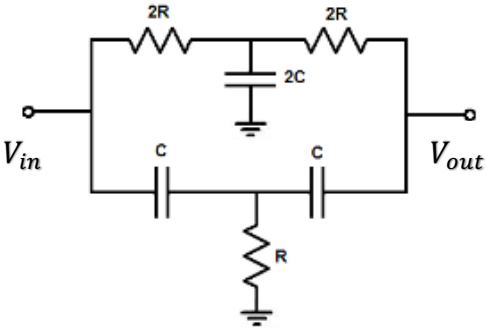
Lab Assignment



Circuit 1



Circuit 2



Circuit 3

1. The lab instructor will assign one circuit to you with appropriate elements values.
2. Connect the circuit using the circuit elements values given by the lab instructor. V_{in} is a sinusoidal AC voltage source with frequency f and maximum amplitude V_{max}.
3. Simulate the circuit using the AC analysis function and generate the transfer function spectrum curve (versus frequency). Use dB scale for the transfer function and log scale for the frequency.
4. From your simulations, find the simulated bandwidth of this filter circuit.
5. Change the value of the circuit elements (one at a time) and observe the simulation curves. Write your conclusions regarding the effect of these elements on the circuit bandwidth. Can you come-up with a formula for the bandwidth in terms of C and R?
6. What kind of a filter is this circuit? Explain.

Experiment (3)
Simulation of Diode Circuits using LTspice

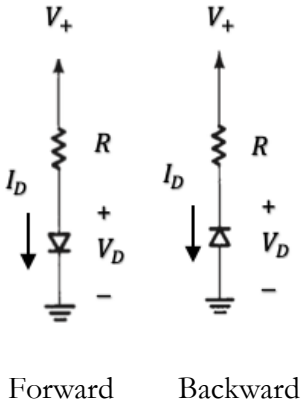
Objectives

- To study the IV characteristics of a standard diode (1N4001) by simulation.
- To simulate simple DC and AC diode circuits.

Pre-lab Work

Analyze the circuits below using the constant voltage diode model (V_{DO}=0.7 V).

Circuits



Forward Backward

Figure 1: Diode IV Circuit.

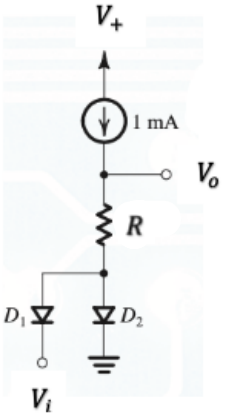


Figure 2: DC diode Circuit.

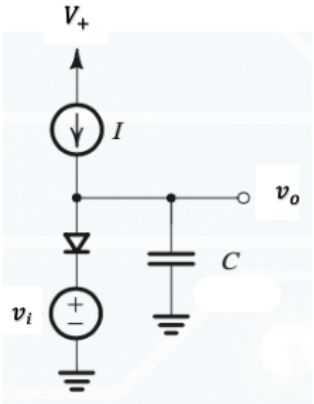


Figure 3: AC Diode Circuit

Procedure

Part (1): Diode IV Characteristics.

1. Connect the circuit in figure 1 (Forward). Use the circuit elements values given by the lab instructor. V_+ is a DC voltage source.
2. Use the DC Sweep function and simulate the response of the circuit to produce the IV curve of the diode (I_D vs. V_D). What is the V_{DO} of this diode? Record the results in table 1.
3. Repeat for the backward circuit. What is the V_{ZO} of this diode? Record the results in table 1.
4. Vary the value of R and repeat the procedure observing the effects (if any).

Part (2): DC Diode Circuit. part2-2.1 dosyasında kayıtlı şema. Direnç ve voltaj değerleri hoca verdi rastgele.Sen değiştirerek ölçümler al. Dc swep değerleri, sırasıyla: V1, linear, 0 , 2, 0.1 olarak aldım

1. Connect the circuit in figure 2. Use the circuit elements values given by the lab instructor. V_+ and V_i are fixed DC voltage sources. diyotların üzerinden geçen akımları 0 yapmaya çalışarak, diyotları on off haline getiricek değerleri denicez.
2. Simulate the circuit. Which diode is ON and which is OFF?

3. With V_i fixed, vary the value of R and repeat step 2 observing the effects on the diodes switching. Record your results in table 2.
4. With R fixed, vary V_i between the range -10 to +10 V and plot the output characteristics curve (V_o vs. V_i).

Part (3): AC Diode Circuit.

1. Connect the circuit in figure 3. Use the circuit elements values given by the lab instructor. I in the circuit is a DC current source and v_i is a sinusoidal signal with small amplitude (less than 10 mV) and a frequency f.
2. Simulate the circuit and plot the waveforms for v_i and v_o (versus time).
3. From the plot in step 2, find the phase shift between v_i and v_o. Record the result in table 3.
4. Vary I and observe the phase shift between v_i and v_o at each setting. Record your results in table 3. Find a value of I that gives a phase shift of $\pm 45^o$?

Results and Analysis

Part (1): Diode IV Characteristics.

Table 1. Diode IV Curve.

R (kΩ)	Simulation V_{DO} (V)	Data sheet V_{DO} (V)	Simulation V_{ZO} (V)	Data sheet V_{ZO} (V)
100				

1. Comment on the results in table 1.

2. Do the IV curves that you obtained resemble the standard diode characteristics? Explain.

Part (2): DC Diode Circuit.

Table 2. Simulated DC results.

R (kΩ)	Diode 1 (ON/OFF)	Diode 2 (ON/OFF)	V_o (V)

1. Comment on the results in table 2.

2. Comment of the output characteristics curve.

Part (3): AC Diode Circuit.

Table 3. Simulated AC results.

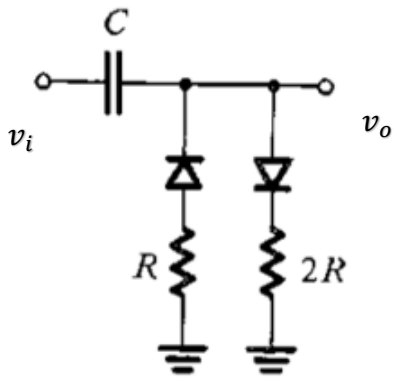
I (mA)	Phase shift

1. Which value of I gives a phase shift of $\pm 45^{o}$?

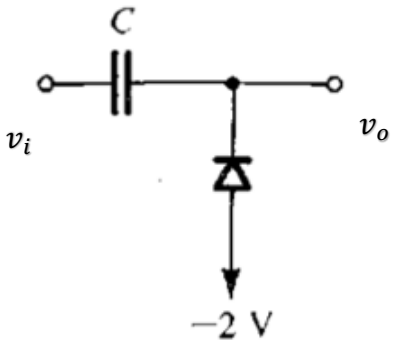
2. Explain the output waveform.

3. Comment on the circuit output characteristics curve.

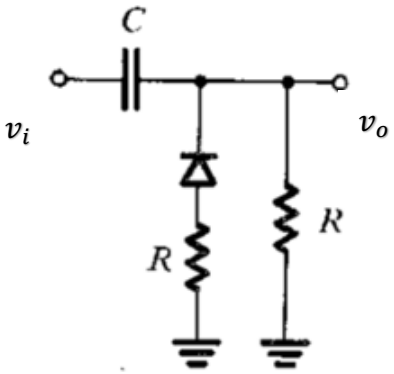
Lab Assignment



Circuit 1

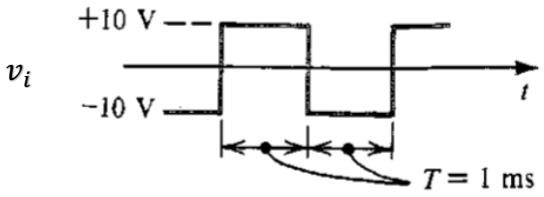


Circuit 2



Circuit 3

1. The lab instructor will assign one circuit to you with appropriate elements values. All diodes are 1N4001 model.
2. Connect the circuit using the circuit elements values given by the lab instructor. v_i is a periodic signal as shown below.
3. Simulate the circuit observing the waveform at the output (v_o).
4. By varying the amplitude of the input signal from 0 to 15 V, produce the transfer function curve of this circuit at the frequency given below.
5. Fix the amplitude at the value below but vary the signal period from 0.1 to 10 ms and make observations regarding the circuit operation.
6. Study the effect of the elements R and C (independently) on the operation of the circuit. Give conclusions.
7. What application can this circuit be good for? Explain.



Name		Name		Name		Name	
ID		ID		ID		ID	
Section							

FAIRCHILD

November 2014

1N4001 - 1N4007
General-Purpose Rectifiers

Features

- Low Forward Voltage Drop
- High Surge Current Capability



DO-41
COLOR BAND DENOTES CATHODE

Ordering Information

Part Number	Top Mark	Package	Packing Method
1N4001	1N4001	DO-204AL (DO-41)	Tape and Reel
1N4002	1N4002	DO-204AL (DO-41)	Tape and Reel
1N4003	1N4003	DO-204AL (DO-41)	Tape and Reel
1N4004	1N4004	DO-204AL (DO-41)	Tape and Reel
1N4005	1N4005	DO-204AL (DO-41)	Tape and Reel
1N4006	1N4006	DO-204AL (DO-41)	Tape and Reel
1N4007	1N4007	DO-204AL (DO-41)	Tape and Reel

Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only. Values are at T_A = 25°C unless otherwise noted.

Symbol	Parameter	Value							Unit
		1N 4001	1N 4002	1N 4003	1N 4004	1N 4005	1N 4006	1N 4007	
V_{RRM}	Peak Repetitive Reverse Voltage	50	100	200	400	600	800	1000	V
$I_{F(AV)}$	Average Rectified Forward Current .375 " Lead Length at T_A = 75°C	1.0							A
I_{FSM}	Non-Repetitive Peak Forward Surge Current 8.3 ms Single Half-Sine-Wave	30							A
I^2t	Rating for Fusing (t < 8.3 ms)	3.7							A^2sec
T_{STG}	Storage Temperature Range	-55 to +175							°C
T_J	Operating Junction Temperature	-55 to +175							°C

Thermal Characteristics

Values are at T_A = 25°C unless otherwise noted.

Symbol	Parameter	Value	Unit
P_D	Power Dissipation	3.0	W
$R_{\theta JA}$	Thermal Resistance, Junction-to-Ambient	50	°C/W

Electrical Characteristics

Values are at T_A = 25°C unless otherwise noted.

Symbol	Parameter	Conditions	Value	Unit
V_F	Forward Voltage	I_F = 1.0 A	1.1	V
I_{rr}	Maximum Full Load Reverse Current, Full Cycle	T_A = 75°C	30	µA
I_R	Reverse Current at Rated V_R	T_A = 25°C	5.0	µA
		T_A = 100°C	50	
C_T	Total Capacitance	V_R = 4.0 V, f = 1.0 MHz	15	pF

Experiment (4)
Diode Characteristics

Objectives

- To experimentally determine the IV characteristics of a general purpose diode.
- To analyze a simple diode circuit using graphical methods.
- To study the application of diodes in logic gate design.

Pre-lab Work

Analyze the two circuits using the constant voltage diode model (V_{DO}=0.7 V).

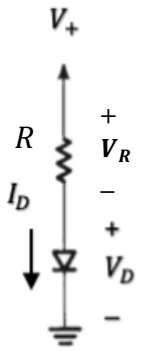


Figure 1: Diode IV curve.

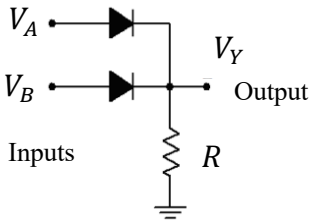


Figure 2: A diode logic gate circuit.

Solution

Equipment

Table 1. Equipment for experiment 4.

	Equipment	Model	Serial No.	Stock No.
1	BreadBoard Station			
2	1 kΩ Resistor (2W, 5%)			
3	Silicon Diodes (2)	1N4001	https://www.onsemi.com/pub/Collateral/1N4001-D.PDF	
4	Multi-meter (1)			
5	Probes and cables			

Experimental Procedure

Part (1): Diode Forward IV Characteristics.

1. Start first by checking if the diode is not damaged. Set the multimeter on resistance measurement. Touch the black lead (negative) of the multimeter to the cathode of the diode (refer to figure 3) and the red lead (positive) to the anode. In this configuration, the diode is forward-biased, and you should get a resistance reading between 1 kΩ and 10 MΩ. Next, switch the leads to the opposite terminals. The diode is now reverse-biased, and the reading should be infinity. If the readings are the same in both directions, the diode is damaged.

2. Connect the circuit in figure 1, with R=1 kΩ. Pay a special attention to how you connect the two terminals of the diode by referring to figure 3.
3. Starting with 0 volts, increase the DC voltage source V_+ in small steps (0.1 V, for example). In each setting, measure the voltage across the diode and across the resistor using the multi-meter. Enter your measurements in table 2.

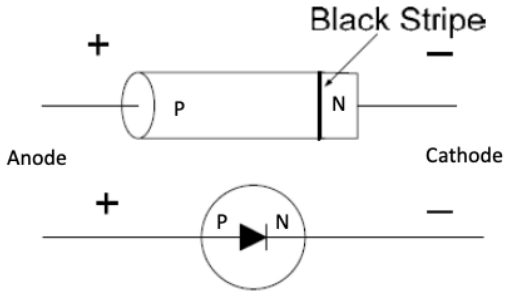


Figure 3: Diode marking.

Part (2): Graphical Analysis of Diode Circuits.

1. Consider again the circuit in figure 1, with a R=1 kΩ. and V_+=2 V.
2. Measure the voltages V_D and V_R using the multi-meter. Enter your measurements in table 3.
3. With the diode removed from the circuit (open-circuit), measure the voltages V_D and V_R. Enter your measurements in table 3.
4. With the diode shorted (a wire is placed instead), measure the voltages V_D and V_R using the multi-meter. Enter measurements in table 3.

Part (3): Diode Application as a Logical Gate.

1. Connect the circuit shown in figure 2.
2. Connect the inputs V_A and V_B to 0 volts or +5V (for a total of four combinations), as shown in table 5. In each combination, record the value of the output voltage V_Y in table 5.

Results and Analysis

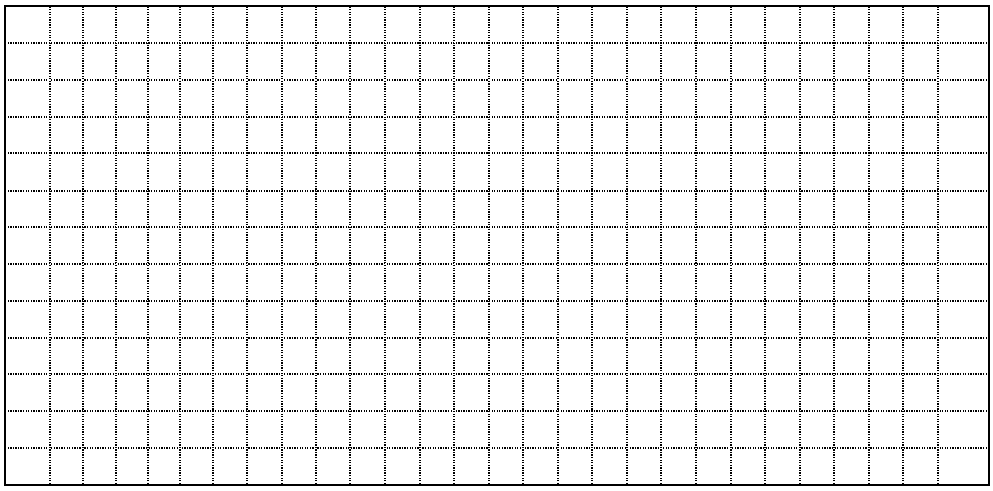
Part (1): Diode Forward IV Characteristics.

1. Calculate the diode current (I_D) in table 2.
2. Now, use the graphing space below to plot the diode characteristics I_D (vertical axis) versus V_R (horizontal axis). Appropriately label your plot.
3. Does the curve resemble the usual diode IV curve? Explain any discrepancies.

Answer:

Table 2. Measurements for Part (1).

DC Input V_+ (V)	V_D (V)	V_R (V)	$I_D = V_R/R$ (mA)



Graph 1: Diode IV characteristics.

Part (2): Graphical Analysis of Diode Circuits.

1. Calculate the diode current (I_D) in table 3 for all three cases.
2. Now, on the same graphing space above (graph 1), draw the "load line" of the circuit. Use the second and third entries in table 3.
3. The intersection point between the load line and the diode characteristics curve is the "operating point" of the circuit. Find this point from the graph and record it in table 4.
4. Also, record in table 4 the experimental current value (table 3, first entry) and the theoretical current value (from pre-lab calculations).
5. Calculate the percent differences. Which value do you think is more accurate? Explain.

Answer:

Table 3. Measurements for Part (2).

Diode	V_D (V)	V_R (V)	$I_D = V_R/R$ (mA)
Included			
Removed			
Shorted			

Table 4. Calculations for Part (2).

I_D (mA) Experimental value	I_D (mA) Graphical value	% Difference	I_D (mA) Theoretical value	% Difference

Part (3): Diode Application as a Logical Gate.

1. From the results in table 5, what logic function does the circuit perform?
2. Diodes are not commonly used in the design of logic gate. Investigate and explain why not?

Answer:

Table 5. Measurements for Part (3).

V_A (V)	V_B (V)	V_Y (V)
0	0	
0	5	
5	0	
5	5	

LTspice Simulations

Include with your report LTspice simulation results of all parts of this experiment.

Conclusions and Remarks

Write below all your conclusions/remarks on this experiment.

Name		Name		Name	
ID		ID		ID	
Section					

FAIRCHILD

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1N4004	1N4004	DO-204AL (DO-41)	Tape and Reel
1N4005	1N4005	DO-204AL (DO-41)	Tape and Reel
1N4006	1N4006	DO-204AL (DO-41)	Tape and Reel
1N4007	1N4007	DO-204AL (DO-41)	Tape and Reel

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Symbol	Parameter	Value							Unit
		1N 4001	1N 4002	1N 4003	1N 4004	1N 4005	1N 4006	1N 4007	
V_{RRM}	Peak Repetitive Reverse Voltage	50	100	200	400	600	800	1000	V
$I_{F(AV)}$	Average Rectified Forward Current .375 " Lead Length at T_A = 75°C	1.0							A
I_{FSM}	Non-Repetitive Peak Forward Surge Current 8.3 ms Single Half-Sine-Wave	30							A
I^2t	Rating for Fusing (t < 8.3 ms)	3.7							A^2sec
T_{STG}	Storage Temperature Range	-55 to +175							°C
T_J	Operating Junction Temperature	-55 to +175							°C

Thermal Characteristics

Values are at T_A = 25°C unless otherwise noted.

Symbol	Parameter	Value	Unit
P_D	Power Dissipation	3.0	W
$R_{\theta JA}$	Thermal Resistance, Junction-to-Ambient	50	°C/W

Electrical Characteristics

Values are at T_A = 25°C unless otherwise noted.

Symbol	Parameter	Conditions	Value	Unit
V_F	Forward Voltage	I_F = 1.0 A	1.1	V
I_{rr}	Maximum Full Load Reverse Current, Full Cycle	T_A = 75°C	30	µA
I_R	Reverse Current at Rated V_R	T_A = 25°C	5.0	µA
		T_A = 100°C	50	
C_T	Total Capacitance	V_R = 4.0 V, f = 1.0 MHz	15	pF

Experiment (5)
Diode Rectifier with Peak Detector

Objectives

- To study the half-wave rectifier function.
- To analyze the output of a peak detector with regulator.

Pre-lab Work

Analyze the circuit below using the constant voltage diode model. The input is a sinusoidal wave with amplitude V_m and frequency f. Find V_o with and without the capacitor and the Zener diode. In each case, draw the waveform of the output and find the average output voltage and average load current.

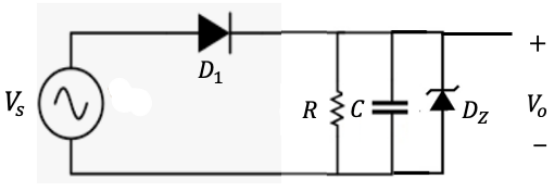


Figure 1: Half-wave rectifier with peak detector and regulator.

Solution

Equipment

Table 1. Equipment for experiment 5.

	Equipment	Model	Serial No.	Stock No.
1	BreadBoard Station			
2	1 kΩ Resistor (2W, 5%)			
3	Capacitors (2)			
4	Diodes (2)	1N4001 (1) 1N4732 (1)		
5	Multi-meter (1)			
6	Dual-trace Oscilloscope			
7	Probes and Cables			

Experimental Procedure

Part (1): Half-wave Rectifier Output Analysis.

1. Connect the circuit in figure 1 without the capacitor and the Zener diode and with R=1 kΩ. Pay a special attention to how you connect the terminals of the diode. The input voltage source V_S is sinusoidal.
2. Connect channel 1 of the oscilloscope to V_S and adjust it to 10 V amplitude at 60 Hz. Pay attention to the polarity of the oscilloscope connection. Sketch the input waveform in graph 1 labeling all important points in both voltage and time axes. Make your sketch as accurate as possible.

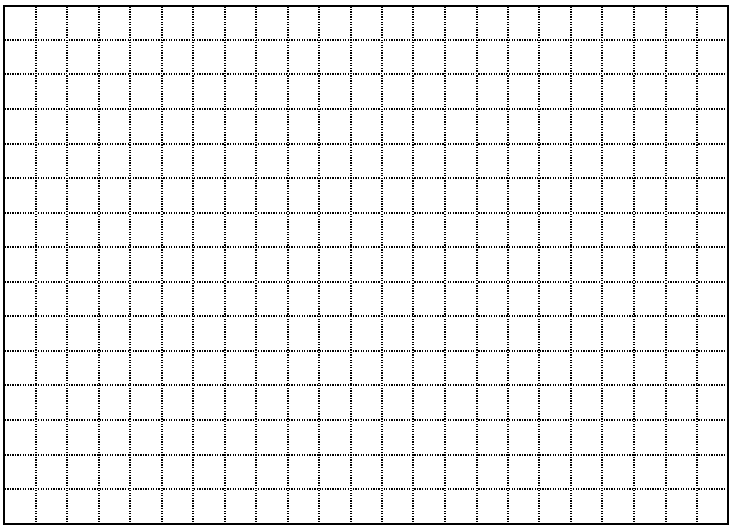
3. Now connect channel 2 of the oscilloscope to V_o. Sketch the output waveform on the same graph 1 labeling all important points in both voltage and time axes. Make your sketch as accurate as possible. Use different colors for the waveforms.
4. Using the multi-meter, measure the output voltage and record it in table 2.

Part (2): Half-wave Rectifier with Peak Detector and Regulator.

1. Consider again the circuit in figure 1 with only the capacitor connected.
2. Repeat the whole procedure in part 1 for both capacitors provided. Record results in graph 1 and table 3. Use different colors for the waveforms.
3. Finally, connect the Zener diode across the capacitor and observe the output. Record results in graph 1. Use different colors for the waveforms.

Results and Analysis

Part (1): Half-wave Rectifier Output Analysis.



Graph 1: Sinusoidal input case.

Table 2. Average output voltage of the half-wave rectifier without peak detector.

Sinusoidal input		
$V_{o\,(ave)}$ (V) Experimental value	$V_{o\,(ave)}$ (V) Theoretical value	% Difference

1. Calculate the theoretical value of the average output voltage and record it in table 2.
2. Comment on your results regarding the waveforms in graph 1.

3. Comment on your results regarding the average output voltage.

Part (2): Half-wave Rectifier with Peak Detector and Regulator.

Table 3. Average output voltage of the half-wave rectifier with peak detector.

	Sinusoidal input		
Capacitor value (μF)	$V_{O\,(ave)}$ (V) Experimental value	$V_{O\,(ave)}$ (V) Theoretical value	% Difference

1. Calculate the theoretical value of the average output voltage and record it in table 3.
2. Comment on your results regarding the waveforms in graph 1.

3. Comment on your results regarding the average output voltage

LTspice Simulations

Include with your report LTspice simulation results of all parts of this experiment.

Conclusions and Remarks

Write below all your conclusions/remarks on this experiment.

Name		Name		Name		Name	
ID		ID		ID		ID	
Section							

Experiment (6)
MOSFET IV Characteristics

Objectives

- To study the IV curve of a MOSFET.
- To graphically find the operating point of a MOSFET circuit.

Pre-lab Work

Analyze the DC circuit below. Assume the transistor is in saturation and find expressions for the drain current and the drain-source voltage. Ignore channel length modulation.

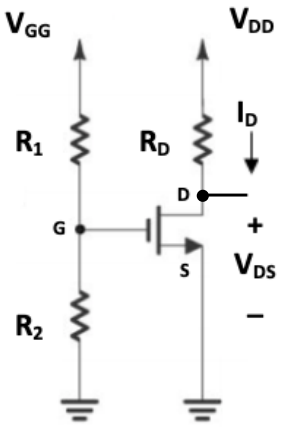


Figure 1: A MOSFET DC Circuit.

Solution

Equipment

Table 1. Equipment for experiment 6.

	Equipment	Model	Serial No.	Stock No.
1	BreadBoard Station			
2	Resistors 1 kΩ (1), 1 MΩ (2)			
3	NMOS transistor (1)	2N7000	https://www.onsemi.com/pub/Collateral/2N7000-D.PDF	
4	Multi-meter (1)			
5	Probes and cables			

Experimental Procedure

Part (1): MOSFET IV Characteristics.

1. Connect the circuit in figure 1 with R_D=1 kΩ and R_1= R_2=1 MΩ. Pay a special attention to how you connect the terminals of the MOSFET (refer to the data sheet to identify the transistor terminals).
2. Set V_{GG} to a constant voltage of +8 V while connecting V_{DD} to a variable DC voltage source.

3. The IV characteristics (I_D versus V_{DS}) is obtained by gradually varying V_{DD} (from 0 to +10 V) and measuring the voltage V_{DS} and V_{R_D} at each setting. Use table 2 to enter your results.
4. Repeat step 3 for another value of V_{GG}, say +10 V. Record your results in table 2.

Part (2): Graphical Analysis of a DC MOSFET Circuit.

1. Reconsider the circuit in figure 1, with $V_{GG} = V_{DD} = +10\,V$. Measure V_{DS} and record the results in table 3. Note that $I_D = (V_{DD} - V_{DS})/R_D$. This is the operating point of this circuit.

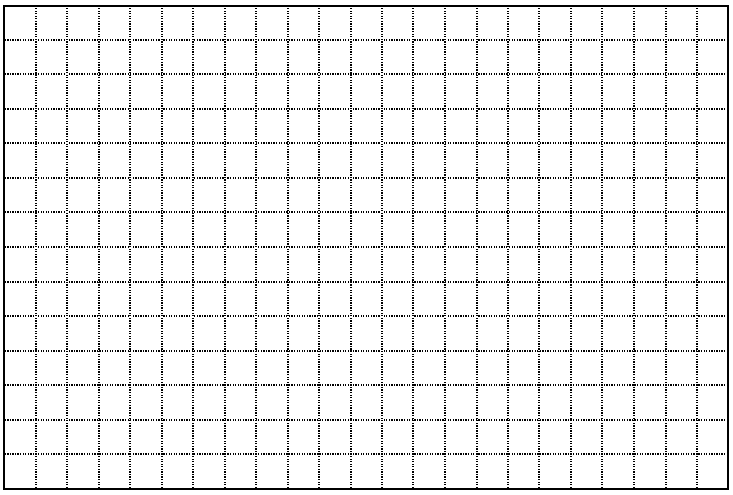
Results and Analysis

Part (1): MOSFET IV Characteristics.

1. Calculate V_{GS} (using voltage divider rule) and the current I_D (V_{R_D}/R_D) in table 2, for both values of V_{GG}.
2. Use graph 1 to sketch the IV curves (I_D versus V_{DS}) for the two V_{GS} values using the results in table 2.

Table 2. IV Characteristics Measurements.

	V_{GG} = 8V, V_{GS} =			V_{GG} = 10V, V_{GS} =		
V_{DD} (V)	V_{DS} (V)	V_{R_D} (V)	I_D (mA)	V_{DS} (V)	V_{R_D} (V)	I_D (mA)
0.0						
1.0						
2.0						
3.0						
4.0						
5.0						
6.0						
7.0						
8.0						
9.0						
10.0						



Graph 1: MOSFET IV Curves.

3. Comment on the IV curves that you obtained. Can you identify the transistor modes of operation?

Answer:

4. Is the drain current constant in the saturation region of the IV curve? Explain why or why not.

Answer:

Part (2): Graphical Analysis of a DC MOSFET Circuit.

1. The load line equation for this circuit is $I_D = \frac{V_{DD}-V_{DS}}{R_D}$. It is the line connecting the two points $(0,\frac{V_{DD}}{R_D})$ and $(V_{DD},0)$ on the IV curve.
2. Draw the load line on top of the IV curves in graph 1. Mark the intersection point of the curves.
3. The intersection of the load line and the IV curve gives the operating point of the circuit. Read the drain current (I_D) value and the drain-source voltage (V_{DS}) value from the intersection point in graph 1. Record the results in table 3.
4. The experimental value of $I_D = (V_{DD} - V_{DS})/R_D$.

Table 3. Graphical Solution Results.

V_{DD}= 10 V and V_{GG} = 10 V					
V_{DS} (V) Experimental value	V_{DS} (V) Graphical value	% Difference	I_D (mA) Experimental value	I_D (mA) Graphical value	% Difference

5. Comment on your results explaining the sources of differences.

Answers:

LTspice Simulations

Include with your report LTspice simulation results of all parts of this experiment.

Conclusions and Remarks

Write below all your conclusions/remarks on this experiment.

Name		Name		Name		Name	
ID		ID		ID		ID	
Section							

2N7000G

Small Signal MOSFET 200 mAmps, 60 Volts

N−Channel TO−92

Features

- AEC Qualified
- PPAP Capable
- This is a Pb−Free Device*

MAXIMUM RATINGS

Rating	Symbol	Value	Unit
Drain Source Voltage	V_{DSS}	60	Vdc
Drain−Gate Voltage (R_{GS} = 1.0 MΩ)	V_{DGR}	60	Vdc
Gate−Source Voltage − Continuous − Non−repetitive ($t_p \le 50$ μs)	 V_{GS} V_{GSM}	 ±20 ±40	 Vdc Vpk
Drain Current − Continuous − Pulsed	 I_D I_{DM}	 200 500	mAdc
Total Power Dissipation @ T_C = 25°C Derate above 25°C	P_D	350 2.8	mW mW/°C
Operating and Storage Temperature Range	T_J, T_{stg}	−55 to +150	°C

THERMAL CHARACTERISTICS

Characteristic	Symbol	Max	Unit
Thermal Resistance, Junction−to−Ambient	$R_{\theta JA}$	357	°C/W
Maximum Lead Temperature for Soldering Purposes, 1/16" from case for 10 seconds	T_L	300	°C

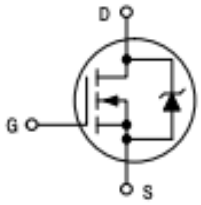
Stresses exceeding Maximum Ratings may damage the device. Maximum Ratings are stress ratings only. Functional operation above the Recommended Operating Conditions is not implied. Extended exposure to stresses above the Recommended Operating Conditions may affect device reliability.

ON Semiconductor®

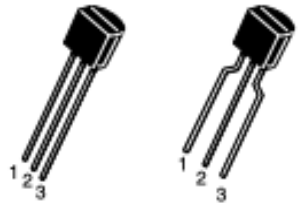
http://onsemi.com

200 mAMPS
60 VOLTS
$R_{DS(on)}$ = 5 Ω

N−Channel



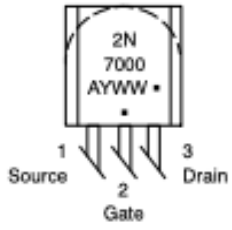
TO−92
CASE 29
STYLE 22



STRAIGHT LEAD BULK PACK

BENT LEAD TAPE & REEL AMMO PACK

MARKING DIAGRAM AND PIN ASSIGNMENT



A = Assembly Location
Y = Year
WW = Work Week
▪ = Pb−Free Package
(Note: Microdot may be in either location)

Experiment (7)
The Common-source Amplifier

Objectives

- To study the operation of a common-source amplifier.
- To verify the voltage gain relations of the common-source amplifier.

Pre-lab Work

Find general expressions for the voltage gain, input resistance and output resistance of the common-source amplifier circuit in figure 1.

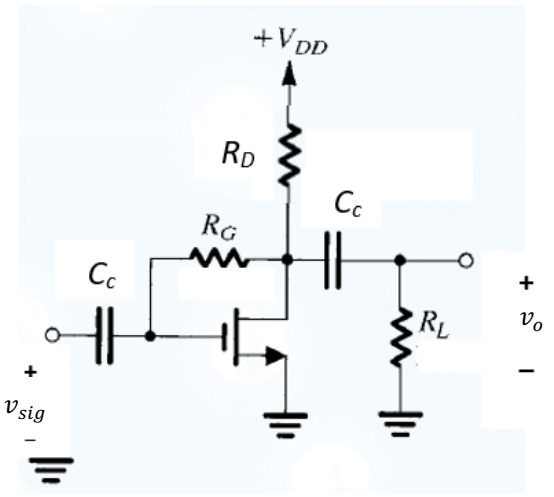


Figure 1: A Common-source Amplifier Circuit.

Solution

Equipment

Table 1. Equipment for experiment 7.

	Equipment	Model	Serial No.	S No.
1	BreadBoard Station			
2	Resistors 10 kΩ (2), 100 kΩ (1), 1 MΩ (1)			
3	NMOS transistor (1)	2N7000	https://www.onsemi.com/pub/Collateral/2N7000-D.PDF	
4	Capacitors 22 μF (2)			
5	Multi-meter (1)			
6	Probes and cables			

Experimental Procedure

Part (1): Amplifier Voltage Gain.

1. Connect the circuit in figure 1 without the ac source, capacitors and the load resistor. Take V_{DD}=10 V, R_D=10 kΩ and R_G=1 MΩ.

2. Measure the DC voltages at the drain and at the gate. Measure the voltage across R_D to find the DC drain current (Use $I_D = \frac{V_{R_D}}{R_D}$). Make sure the transistor is operating in saturation mode.
3. Now add the capacitors and the load resistor (R_L=10 kΩ) and connect a sinusoidal ac source v_{sig}. Start with a small amplitude (in mV) at 10 kHz for v_{sig}.
4. Connect both the input signal and the output terminals to the oscilloscope using two channels.
5. Observe both waveforms simultaneously and produce a sketch using graph 1.
6. Increase the amplitude of v_{sig} in steps of a few mV. In each case record the amplitudes of v_{sig} and v_o in table 2.
7. Continue until you notice that the output becomes distorted. Mark the maximum input and output before distortion.

Part (2): Input and Output Resistance of Amplifier.

1. Reconsider the circuit in figure 1. Set the input signal at a comfortable value within the linear range of the amplifier.
2. To find the output resistance of the amplifier, measure the voltage gain with and without the load resistor. Record results in table 3.
3. To find the input resistance, connect a resistor (R_x=100 kΩ) between the input voltage source and the coupling capacitor (in series). Measure the voltage gain with and without R_x. Record results in table 3.

Results and Analysis

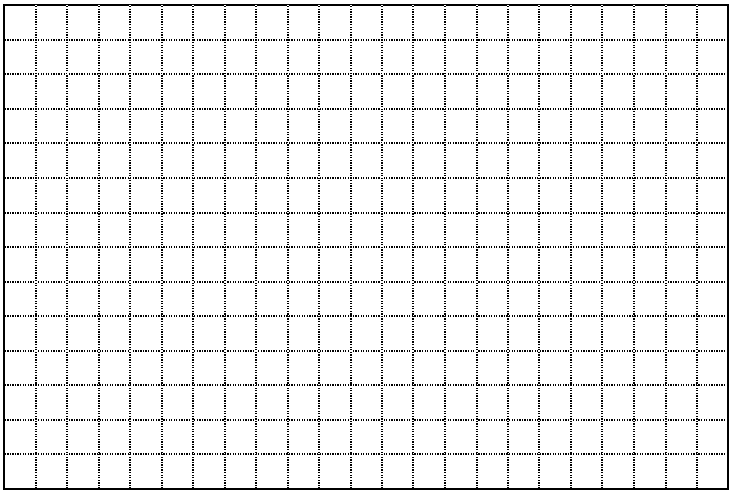
Part (1): Amplifier Voltage Gain.

1. Calculate the voltage gain of this amplifier (v_o/v_{sig}) and record your results in table 2.
2. Comment on the linearity of the amplifier in this circuit.

Answer:

3. What is the maximum output swing for this set-up of the amplifier? Explain.

Answer:



Graph 1: Amplifier input/output waveforms.

Table 2. Voltage Gain Results

Amplitudes		
v_{sig} (mV)	v_o (mV)	Gain

Part (2): Input and Output Resistance of Amplifier.

1. Calculate the input and output resistances of this amplifier circuit and record them in table 3.

Table 3. Input/output Resistances.

Output Resistance (kΩ)			Input Resistance (kΩ)		
Gain with R_L	Gain without R_L	R_o	Gain with R_x	Gain without R_x	R_i

2. Comment on your results based on your knowledge of the properties of the common-source amplifier.

Answer:

LTspice Simulations

Include with your report LTspice simulation results of all parts of this experiment.

Conclusions and Remarks

Write below all your conclusions/remarks on this experiment.

Name		Name		Name		Name	
ID		ID		ID		ID	
Section							

Experiment (8)
BJT IV Characteristics

Objectives

- To study the IV curve of a BJT.
- To experimentally evaluate the DC gain parameter of a BJT.
- To study the transfer function of a BJT circuit.

Pre-lab Work

Analyze the DC circuit below. Assume the transistor is in active mode and find expressions for the collector current and the collector-emitter voltage.

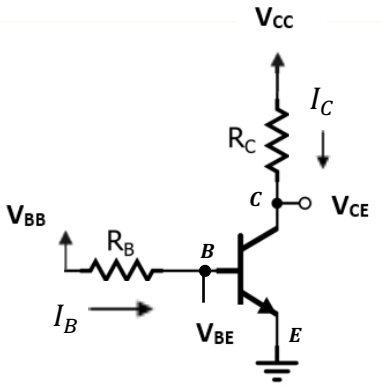


Figure 1: A BJT DC Circuit.

Solution

Equipment

Table 1. Equipment for experiment 8.

	Equipment	Model	Serial No.	Stock No.
1	BreadBoard Station			
2	Resistors 5 kΩ (1), 10 kΩ (1), 100 kΩ (1)			
3	BJT NPN transistor (1)	2N3904	https://www.onsemi.com/pub/Collateral/2N3903-D.PDF	
4	Multi-meter (1)			
5	Probes and cables			

Experimental Procedure

Part (1): BJT IV Characteristics

1. Connect the circuit in figure 1 with R_C=10 kΩ and R_B=100 kΩ. Pay a special attention to how you connect the terminals of the BJT. Refer to the data sheet to identify the transistor terminals.

2. Set V_{BB} to a constant voltage of +10 V while connecting V_{CC} to a variable DC voltage source.
3. The IV characteristics (I_C versus V_{CE}) is obtained by slowly varying V_{CC} and measuring the voltage V_{CE} and V_{R_C} at each setting. Use table 2 to enter your results. Also record the measured values of V_{R_B} at each setting (to be used in part 2 of the experiment).

Part (2): BJT DC Gain Parameter (β)

1. The required data have been collected in part 1.

Part (3): BJT Transfer Function

1. Reconsider the circuit in figure 1.
2. Set V_{CC} to a constant voltage of +10 V while connecting V_{BB} to a variable DC voltage source.
3. The transfer function (V_{CE} versus V_{BB}) is obtained by slowly varying V_{BB} and measuring the voltage V_{CE} at each setting. Use table 3 to enter your results.
4. Repeat step 3 for another value of R_C, say 5 kΩ. Record your results in table 3.

Results and Analysis

Part (1): BJT IV Characteristics.

1. Calculate the currents I_C and I_B values in table 2 (Use $I_C = V_{RC}/R_C$ and $I_B = V_{RB}/R_B$).
2. Use graph 1 to sketch the IV curve.

Table 2. IV Characteristics Measurements.

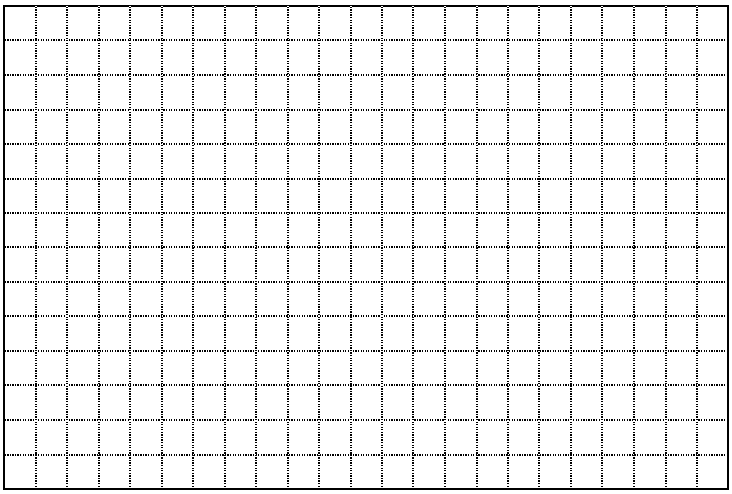
V_{CC} (V)	V_{CE} (V)	V_{R_C} (V)	I_C (mA)	V_{R_B} (V)	I_B (mA)	$\beta = I_C/I_B$
0.0						

3. Comment on the IV curve that you obtained. Can you identify the transistor modes of operation? Show your answer on the curve.

Answer:

4. Is the collector current constant in the active mode region of the IV curve? Find an approximate value of the Early voltage.

Answer:



Graph 1: Transistor IV Curve.

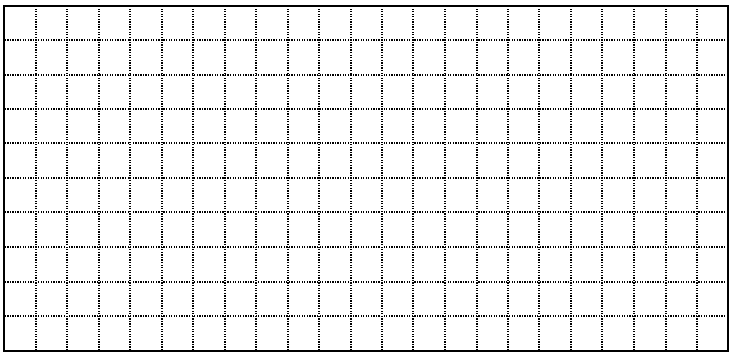
Part (2): BJT DC Gain Parameter (β)

1. Calculate the current I_B values in table 2. Also, calculate the DC gain parameter (β).
2. Why is the value of β not constant in the table? Explain in details.

Answer:

3. Use graph 2 to sketch β vs. I_C relation. Compare to the curve given in the transistor data sheet and give your comments.

Answer:



Graph 2: DC gain parameter.

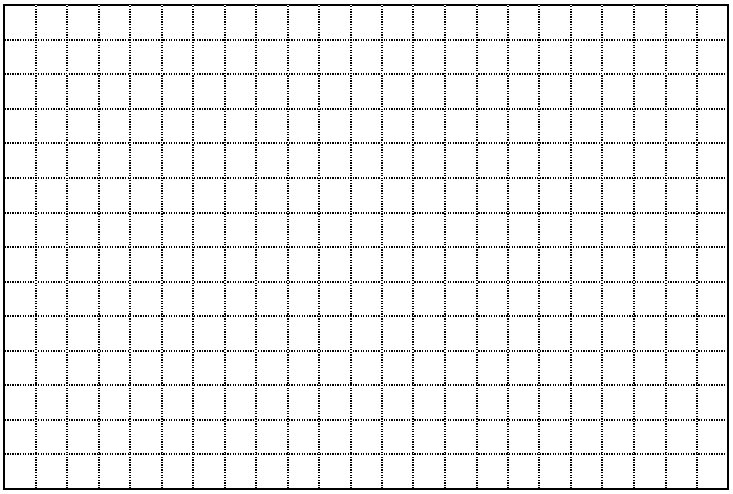
Part (3): BJT Transfer Function

1. Use graph 3 to sketch the two transfer functions using the data in table 3, for both R_C values.
2. Can you identify the transistor modes of operation? Show your answer on the curves.

Answer:

Table 3. Transfer Function Measurements.

	R_C= 10 kΩ	R_C= 5 kΩ
V_{BB} (V)	V_{CE} (V)	V_{CE} (V)
0.0		



Graph 3: Circuit transfer function.

3. Comment on the general shape of the transfer function. In particular, explain the main differences between the two curves, in relation to R_C value.

Answer:

LTspice Simulations

Include with your report LTspice simulation results of all parts of this experiment.

Conclusions and Remarks

Write below all your conclusions/remarks on this experiment.

Name		Name		Name		Name	
ID		ID		ID		ID	
Section							

2N3904
Silicon NPN Transistor
General Purpose
TO–92 Type Package

Absolute Maximum Ratings:

Collector–Emitter Voltage, V_{CEO} 40V
Collector–Base Voltage, V_{CB} 60V
Emitter–Base Voltage, V_{EBO} 6V
Continuous Collector Current, I_C 200mA
Total Device Dissipation (T_A = +25°C), P_D 625mW
Derate Above 25°C . . . . . 2.8mW/°C
Total Device Dissipation (T_C = +25°C), P_D 1.5W
Derate Above 25°C . . . . . 12mW/°C
Operating Junction Temperature Range, T_J . . . . . –55° to +150°C
Storage Temperature Range, T_{stg} . . . . . –55° to +150°C
Thermal Resistance, Junction to Case, R_{thJC} . . . . . 83.3°C/W
Thermal Resistance, Junction to Ambient, R_{thJA} . . . . . 200°C/W

Electrical Characteristics: (T_A = +25°C unless otherwise specified)

Parameter	Symbol	Test Conditions	Min	Typ	Max	Unit
OFF Characteristics						
Collector–Emitter Breakdown Voltage	$V_{(BR)CEO}$	I_C = 1mA, I_B = 0, Note 1	40	–	–	V
Collector–Base Breakdown Voltage	$V_{(BR)CBO}$	I_C = 10μA, I_E = 0	60	–	–	V
Emitter–Base Breakdown Voltage	$V_{(BR)EBO}$	I_E = 10μA, I_C = 0	6	–	–	V
Collector Cutoff Current	I_{CEX}	V_{CE} = 30V, V_{EB} = 3V	–	–	50	nA
Base Cutoff Current	I_{BL}	V_{CE} = 30V, V_{EB} = 3V	–	–	50	nA
ON Characteristics (Note 1)						
DC Current Gain	h_{FE}	V_{CE} = 1V, I_C = 0.1mA	40	–	–	
		V_{CE} = 1V, I_C = 1mA	70	–	–	
		V_{CE} = 1V, I_C = 10mA	100	–	300	
		V_{CE} = 1V, I_C = 50mA	60	–	–	
		V_{CE} = 1V, I_C = 100mA	30	–	–	

Note 1. Pulse Test: Pulse Width ≤ 300μs, Duty Cycle ≤ 2%.

Electrical Characteristics (Cont'd): (T_A = +25°C unless otherwise specified)

Parameter	Symbol	Test Conditions	Min	Typ	Max	Unit
ON Characteristics (Cont'd) (Note 1)						
Collector–Emitter Saturation Voltage	$V_{CE(sat)}$	I_C = 10mA, I_B = 1mA	–	–	0.2	V
		I_C = 50mA, I_B = 5mA	–	–	0.3	V
Base–Emitter Saturation Voltage	$V_{BE(sat)}$	I_C = 10mA, I_B = 1mA	0.65	–	0.85	V
		I_C = 50mA, I_B = 5mA	–	–	0.95	V
Small–Signal Characteristics						
Current Gain–Bandwidth Product	f_T	I_C = 10mA, V_{CE} = 20V, f = 100MHz	300	–	–	MHz
Output Capacitance	C_{obo}	V_{CB} = 5V, I_E = 0, f = 1MHz	–	–	4.0	pF
Input Capacitance	C_{ibo}	V_{CB} = 0.5V, I_C = 0, f = 1MHz	–	–	8.0	pF
Input Impedance	h_{ie}	I_C = 1mA, V_{CE} = 10V, f = 1kHz	1.0	–	10	kΩ
Voltage Feedback Ratio	h_{re}	I_C = 1mA, V_{CE} = 10V, f = 1kHz	0.5	–	8.0	x 10^{-4}
Small–Signal Current Gain	h_{fe}	I_C = 1mA, V_{CE} = 10V, f = 1kHz	100	–	400	
Output Admittance	h_{oe}	I_C = 1mA, V_{CE} = 10V, f = 1kHz	1.0	–	30	μmhos
Noise Figure	NF	I_C = 100μA, V_{CE} = 5V, R_S = 1kΩ, f = 10Hz to 15.7kHz	–	–	5.0	db
Switching Characteristics						
Delay Time	t_d	V_{CC} = 3V, V_{EB} = 0.5V, I_C = 10mA, I_{B1} = 1mA	–	–	35	ns
Rise Time	t_r		–	–	35	ns
Storage Time	t_s	V_{CC} = 3V, I_C = 10mA, I_{B1} = I_{B2} = 1mA	–	–	200	ns
Fall Time	t_f		–	–	50	ns

Note 1. Pulse Test: Pulse Width ≤ 300μs, Duty Cycle ≤ 2%.

COMMENTS AND REMARKS ON THE LAB MANUAL